

ENUNCIADO — PROBLEMA 1

Numerical Analysis – Preliminary Exam

Department of Mathematical Sciences

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Instructions: Answer all four of the following problems. This is a closed book, closed notes exam. Show all of your work for full credit. No calculators or computers are allowed.

1. One iterative method for solving the linear system $A\mathbf{x} = \mathbf{b}$ [A is $n \times n$, $\mathbf{x}$ and $\mathbf{b}$ are $n \times 1$] has the form

$$\mathbf{x}_{k+1} = G\mathbf{x}_k + \mathbf{c}, \quad \text{for } k = 0, 1, 2, \dots$$

with $G = M^{-1}N$ and $\mathbf{c} = M^{-1}\mathbf{b}$ where the matrices M and N come from a splitting of the matrix $A = M - N$ with M nonsingular. The iterative scheme is started with an initial guess $\mathbf{x}_0$.

(a) Under what conditions on the matrix G will this iterative scheme converge to a solution of $A\mathbf{x} = \mathbf{b}$? (You may assume that G is a nondefective matrix). Be sure to justify your answer.

(b) The Jacobi method uses $M = D$ and $N = -(L+U)$ where D is a diagonal matrix with the same diagonal elements of A while L and U are strictly lower and upper triangular matrices, respectively so that $A = D + L + U$. The iterative scheme is

$$\mathbf{x}_{k+1} = -D^{-1}(L+U)\mathbf{x}_k + D^{-1}\mathbf{b}$$

A modification of this method is the Gauss-Seidel method in which the iterative scheme is given by

$$\mathbf{x}_{k+1} = -D^{-1}L\mathbf{x}_{k+1} - D^{-1}U\mathbf{x}_k + D^{-1}\mathbf{b}$$

where D , L and U are defined as above. Explain and/or interpret as many similarities and differences between these two methods that you can identify, including implementation details.

2. Consider the overdetermined system $A\mathbf{x} = \mathbf{b}$ where A is an $m \times n$ matrix of full rank, $\mathbf{x}$ is a vector of length n and $\mathbf{b}$ is a vector of length m . Here $m > n$. A weighted least squares problem seeks $\mathbf{x}$ such that the norm

$$\|D(\mathbf{b} - A\mathbf{x})\|_2$$

is minimized. Here D is a full rank $m \times m$ diagonal matrix with entries d_{jj} [The case $D = I$ corresponds to the non-weighted least squares problem.]

- (a) Write down the corresponding normal equations for the weighted least squares problem in terms of the matrices A and D and the vectors $\mathbf{x}$ and $\mathbf{b}$.
- (b) In the standard (non-weighted) least squares problem another interpretation of the solution $\mathbf{x}$ is that it solves the equation $A\mathbf{x} = P\mathbf{b}$ where P is an orthogonal projector onto the range (or span) of A . Give a formula for the analogous orthogonal projector P_W for the weighted least squares problem.
3. Consider Heun's method for solving ODEs given by $x_{k+1} = x_k + \frac{h_k}{2}(k_1 + k_2)$, where $k_1 = f(t_k, x_k)$, $k_2 = f(t_k + h_k, x_k + h_k k_1)$.
- (a) Determine the order of accuracy and the stability region for this method.
- (b) Characterize the method as implicit/explicit, single-step or multistep.
- (c) Define what it means for the ODE to be considered stiff. Suggest a possible modification of Heun's method to make it more suitable for the solution of stiff equations. Explain.
4. (a) Given the three data points $(-1, 1), (0, 0), (1, 1)$ determine the interpolating polynomial of degree two using monomial basis, Lagrange basis and Newton basis.
- (b) Rank the three methods above according to the cost of determining the interpolant, from least to most expensive. Which of the three methods above has the best-conditioned basis matrix?

SOLUCIÓN — PROBLEMA 1

Problema 1 — Convergencia del esquema iterativo; Jacobi vs. Gauss-Seidel

(a) El esquema $x^{(k+1)} = Gx^{(k)} + c$ con $e^{(k)} = x^{(k)} - x^*$ satisface $e^{(k+1)} = Ge^{(k)}$, así que $e^{(k)} = G^k e^{(0)}$. Como G es no defectiva, existe una base de vectores propios $\{v_i\}$ con $Gv_i = \lambda_i v_i$; en esa base $G^k e^{(0)} = \sum_i c_i \lambda_i^k v_i \rightarrow 0$ para todo $e^{(0)}$ si y solo si $|\lambda_i| < 1$ para todo i , es decir $\rho(G) < 1$.

Recíprocamente, si $\rho(G) \geq 1$ existe $|\lambda| \geq 1$ con vector propio v ; tomando $e^{(0)} = v$ da $e^{(k)} = \lambda^k v \not\rightarrow 0$. Luego la condición necesaria y suficiente es $\boxed{\rho(G) < 1}$.

(b) Comparación Jacobi vs. Gauss-Seidel:

Aspecto	Jacobi	Gauss-Seidel
Iteración	$x_i^{(k+1)} = (b_i - \sum_{j \neq i} a_{ij} x_j^{(k)}) / a_{ii}$	$x_i^{(k+1)} = (b_i - \sum_{j < i} a_{ij} x_j^{(k+1)} - \sum_{j > i} a_{ij} x_j^{(k)}) / a_{ii}$
Uso de nuevos valores	No (usa solo $x^{(k)}$)	Sí (usa $x^{(k+1)}$ ya disponibles)
Paralelismo	Completamente paralelizable	No paralelizable en la forma estándar
Almacenamiento	Requiere dos vectores ($x^{(k)}$ y $x^{(k+1)}$)	Puede hacerse in-place con un solo vector
Convergencia	$G_J = -D^{-1}(L + U)$, $\rho(G_J)$ puede ser mayor o menor	$G_{GS} = -(D + L)^{-1}U$; en general $\rho(G_{GS}) \leq \rho(G_J)^2$ para matrices tridiagonales
Cuándo converge	Para A diagonal dominante estricta por filas (y otros casos)	Bajo las mismas condiciones, con frecuencia más rápido
Implementación	Más simple (todos los $x_j^{(k)}$ disponibles al inicio)	Actualización secuencial (forward substitution)

Ambos son iteraciones tipo punto fijo con matrices de iteración distintas. Para A estrictamente diagonalmente dominante por filas, ambos convergen; Gauss-Seidel converge típicamente en la mitad de iteraciones de Jacobi (en ese caso, $\rho(G_{GS}) = \rho(G_J)^2$).

ENUNCIADO — PROBLEMA 2

2. Consider the overdetermined system $A\mathbf{x} = \mathbf{b}$ where A is an $m \times n$ matrix of full rank, $\mathbf{x}$ is a vector of length n and $\mathbf{b}$ is a vector of length m . Here $m > n$. A weighted least squares problem seeks $\mathbf{x}$ such that the norm

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- (a) Write down the corresponding normal equations for the weighted least squares problem in terms of the matrices A and D and the vectors $\mathbf{x}$ and $\mathbf{b}$.
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Problema 2 — Mínimos cuadrados ponderados

(a) Minimizar $\|D(b - Ax)\|_2^2 = (b - Ax)^T D^T D (b - Ax) = (b - Ax)^T W (b - Ax)$ donde $W = D^T D$ (simétrica y definida positiva porque D es de rango completo). El gradiente con respecto a x es

$$\nabla_x [(b - Ax)^T W (b - Ax)] = -2A^T W (b - Ax) = 0.$$

Las ecuaciones normales ponderadas son

$$(A^T W A)x = A^T W b, \quad W = D^T D.$$

Si D es diagonal con entradas d_{jj} , entonces $W = \text{diag}(d_{jj}^2)$.

(b) En mínimos cuadrados estándar la solución satisface $Ax = Pb$ con $P = A(A^T A)^{-1} A^T$ (proyector ortogonal sobre $\text{ran}(A)$). Para el caso ponderado, la solución de $(A^T W A)x = A^T W b$ satisface $Ax = P_W b$ donde el proyector ortogonal **en la norma $\|\cdot\|_W$ inducida por** W es

$$P_W = A(A^T W A)^{-1} A^T W.$$

Verificación: $P_W A = A(A^T W A)^{-1} A^T W A = A$ (proyecta vectores en $\text{ran}(A)$ sobre sí mismos), y $P_W^2 = A(A^T W A)^{-1} A^T W \cdot A(A^T W A)^{-1} A^T W = P_W$ (idempotente). P_W es W -autoadjunto: $\langle P_W u, v \rangle_W = \langle u, P_W v \rangle_W$ (puede verificarse directamente con la fórmula).

ENUNCIADO — PROBLEMA 3

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Problema 3 — Método de Heun para EDO

(a) El orden de exactitud y la región de estabilidad se determinaron en el examen 2012-01 problema 3 y también en 2011-01 problema 4. Aquí el método es idéntico ($x_{k+1} = x_k + \frac{h_k}{2}(k_1 + k_2)$, $k_1 = f(t_k, x_k)$, $k_2 = f(t_k + h_k, x_k + h_k k_1)$).

Aplicando a $x' = \lambda x$: factor de amplificación $R(z) = 1 + z + \frac{1}{2}z^2$ con $z = h_k \lambda$. Coincide con e^z hasta orden 2 (error $O(z^3)$): **método de orden 2**. Región de estabilidad real ($\lambda < 0$): $|R(z)| \leq 1$ para $-2 < z \leq 0$, es decir $h_k \leq 2/|\lambda|$.

(b) El método de Heun es **explícito** (ambas evaluaciones de f usan valores ya conocidos), **de un solo paso** (solo involucra el nivel k para calcular el nivel $k + 1$), y de **dos etapas** (Runge-Kutta de orden 2).

(c) Una EDO $x' = f(t, x)$ es **rígida** (stiff) cuando tiene componentes con constantes de tiempo muy distintas: el Jacobiano $\partial f / \partial x$ tiene valores propios con partes reales muy negativas de magnitudes muy distintas. En la práctica esto obliga a usar pasos h muy pequeños para la estabilidad del método explícito aunque la precisión no lo requiera.

Modificación de Heun para rigidez: reemplazar la evaluación explícita de k_2 por una implícita (método trapezoidal / Euler implícito en el segundo estadio):

$$k_2 = f(t_k + h_k, x_k + h_k k_2) \quad (\text{solve implicitly for } k_2),$$

dando el **método trapezoidal implícito** (o Crank-Nicolson para EDOs). Éste es A-estable (incondicionalmente estable para $\lambda < 0$), con la misma precisión de orden 2, y mucho más adecuado para problemas rígidos.

ENUNCIADO — PROBLEMA 4

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(c) Derive the formula for the interpolation error $f(x) - p(x)$, where $p(x)$ is a polynomial of degree at most n that interpolates f at $n + 1$ distinct nodes $x_0, x_1, \dots, x_n$ on the interval $[a, b]$ where f is sufficiently smooth.

(d) Find the interpolation error associated with the interpolants found in (a) on the interval $[-1, 1]$ assuming the function being interpolated is given by $f(x) = x^4$.

SOLUCIÓN — PROBLEMA 4

Problema 4 — Interpolación polinómica: tres bases, error

(a) Datos: $(-1, 1), (0, 0), (1, 1)$. El polinomio buscado es $p(x)$ de grado ≤ 2 .

Base monomio: $p(x) = c_0 + c_1x + c_2x^2$. Sistema de Vandermonde:

$$\begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

De la fila 2: $c_0 = 0$. Las filas 1 y 3: $-c_1 + c_2 = 1$ y $c_1 + c_2 = 1$, dando $c_2 = 1, c_1 = 0$.

$$\boxed{p(x) = x^2.}$$

Base de Lagrange: $p(x) = y_0L_0(x) + y_1L_1(x) + y_2L_2(x)$ con $y_0 = 1, y_1 = 0, y_2 = 1$:

$$L_0(x) = \frac{(x-0)(x-1)}{(-1-0)(-1-1)} = \frac{x(x-1)}{2}, \quad L_1(x) = \frac{(x+1)(x-1)}{(0+1)(0-1)} = -(x^2-1),$$

$$L_2(x) = \frac{(x+1)(x-0)}{(1+1)(1-0)} = \frac{x(x+1)}{2}.$$

$$p(x) = 1 \cdot \frac{x(x-1)}{2} + 0 + 1 \cdot \frac{x(x+1)}{2} = \frac{x^2 - x + x^2 + x}{2} = x^2. \checkmark$$

Base de Newton (diferencias divididas):

x_i	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_0, x_1, x_2]$
-1	1	$(0-1)/(0-(-1)) = -1$	$(1-(-1))/(1-(-1)) = 1$
0	0	$(1-0)/(1-0) = 1$	
1	1		

$$p(x) = f[-1] + (x+1)f[-1, 0] + (x+1)(x-0)f[-1, 0, 1] = 1 + (x+1)(-1) + (x+1)(x)(1) = 1 - x - 1 + x^2 + x = x^2. \checkmark$$

(b) Costo y condicionamiento:

1. **Newton** es el menos caro para calcular el interpolante (diferencias divididas: $O(n^2)$ operaciones, compactas y actualizables al agregar nodos).
2. **Lagrange** tiene costo similar $O(n^2)$ pero es más costoso para evaluar o agregar nodos.
3. **Monomio (Vandermonde)** es el más caro: resolver el sistema de Vandermonde cuesta $O(n^3)$ y la matriz de Vandermonde tiene **el peor condicionamiento** (número de condición exponencial en n).

La base mejor condicionada es la de **Newton** (las diferencias divididas están bien condicionadas numéricamente), seguida de Lagrange; la base monomio tiene el peor condicionamiento.

(c) Formula del error de interpolación (derivación por función auxiliar):

Sea $p(x)$ el interpolante en $n + 1$ nodos distintos $x_0, \dots, x_n \in [a, b]$ con $f \in C^{n+1}[a, b]$. Para $x \neq x_i$ definimos $\omega(t) = \prod_{i=0}^n (t - x_i)$ y la función auxiliar

$$\varphi(t) = f(t) - p(t) - \frac{f(x) - p(x)}{\omega(x)} \omega(t).$$

φ se anula en los $n + 2$ puntos $x_0, \dots, x_n, x$. Por el teorema de Rolle aplicado $n + 1$ veces, existe $\xi = \xi(x) \in (a, b)$ con $\varphi^{(n+1)}(\xi) = 0$. Como $p^{(n+1)} \equiv 0$ y $\omega^{(n+1)} \equiv (n + 1)!$:

$$f^{(n+1)}(\xi) - \frac{f(x) - p(x)}{\omega(x)} (n + 1)! = 0 \implies f(x) - p(x) = \frac{\omega(x)}{(n + 1)!} f^{(n+1)}(\xi). \blacksquare$$

(d) Para $f(x) = x^4$, $n = 2$ ($n + 1 = 3$ nodos $-1, 0, 1$), $f^{(3)}(x) = 24x$, y $\omega(x) = (x + 1)(x)(x - 1) = x^3 - x$. El error es

$$f(x) - p(x) = \frac{(x^3 - x)}{3!} f^{(3)}(\xi) = \frac{(x^3 - x) \cdot 24\xi}{6} = 4\xi(x^3 - x),$$

pero como $p(x) = x^2$, $f(x) - p(x) = x^4 - x^2$ exactamente. El error máximo en $[-1, 1]$: $|x^4 - x^2|$ se maximiza en $x = \pm 1/\sqrt{2}$, dando $|1/4 - 1/2| = 1/4$.